MBTI Prediction & Analysis

Abstract

Personality prediction from social media text has become a vital area in computational psychology. This study investigates the linguistic patterns of Twitter users and evaluates the effectiveness of deep learning models in predicting MBTI personality types. Utilizing a dataset of over 7,000 labeled users, we conducted a comprehensive analysis combining TF-IDF, lexical-based sentiment matching, and TextCNN classification.

Our exploratory analysis reveals a divergence between social media personality distributions and global norms, specifically an underrepresentation of "ES" types. Linguistic analysis indicates that "Feeling" types employ significantly more positive emotion words compared to "Thinking" types, and specific keywords (e.g., "artwork", "raffle") show strong cross-dimensional correlations with personality traits. For classification, the TextCNN model achieved an accuracy of approximately 80% on a filtered dataset, though it demonstrated sensitivity to noise and overfitting risks. These findings confirm that while language is a powerful predictor of personality, robust modeling requires careful data balancing and noise management.

Keywords: MBTI, Personality Prediction, TextCNN, Twitter Data, Natural Language Processing

1 Introduction

The Myers-Briggs Type Indicator (MBTI) has emerged as a significant psychological instrument for personality assessment, widely utilized in career planning, team management, and self-cognition. While traditional MBTI assessments rely on structured questionnaires, they are often time-consuming and susceptible to subjective bias. In the digital age, social media platforms like Twitter provide a vast repository of spontaneous user-generated content, offering a novel avenue for implicit personality prediction through natural language processing (NLP) techniques.

This study aims to bridge the gap between psychological typing and computational linguistics by analyzing linguistic patterns in Twitter data. Unlike traditional psychological studies that rely on small-scale surveys, we leverage a large-scale dataset from Kaggle to investigate whether specific word usage patterns and discourse styles correlate with MBTI dimensions.

Our research reveals a distinct divergence between the user distribution in our Twitter dataset and the general population norms reported in the 16Personalities World Personality Map. Specifically, we observed an underrepresentation of the "ES" cluster (ESTJ, ESTP, ESFP, ESFJ), suggesting that Twitter users may possess distinct personality traits compared to the broader demographic. To address the challenges of text classification in this unbalanced dataset, we employ deep learning architectures, specifically the TextCNN model proposed by Yoon Kim, to extract high-dimensional semantic features for personality prediction.

2 Related Work

2.1 Personality Distribution Analysis

Prior research on personality distribution often relies on standardized baselines. In this study, we utilize the 16Personalities World Personality Map (Fourth Edition) as a benchmark to assess the representativeness of our dataset. Comparative analysis indicates that while ISTJ is the most common type globally, social media datasets often exhibit skewness towards Introverted (I) and Intuitive (N) types, necessitating careful data balancing strategies during model training.

2.2 Text Classification and Deep Learning

In the field of Natural Language Processing (NLP), Convolutional Neural Networks (CNN) have demonstrated remarkable efficacy in sentence-level classification tasks. The TextCNN model, introduced by Yoon Kim (2014), utilizes convolutional filters of varying window sizes to capture local semantic features (n-grams) effectively. This architecture is particularly suitable for MBTI classification tasks where specific linguistic cues (e.g., emotional keywords or logical connectives) play a decisive role in distinguishing between "Feeling" (F) and "Thinking" (T) types.

2.3 Lexical Analysis

Beyond deep learning, lexical-based approaches provide interpretability for personality prediction. Previous studies have focused on lexicon matching and word frequency analysis to quantify linguistic differences. In our methodology, we extend this by integrating Term Frequency-Inverse Document Frequency (TF-IDF) and Word Cloud visualization to explicitly map the vocabulary preferences of different MBTI types, such as the distinct divergence between ISTP and ENFJ discourse patterns.

3 Methodology

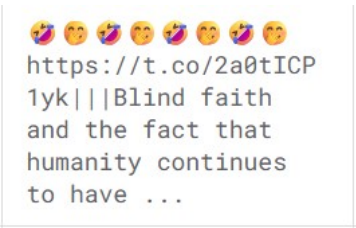
3.1 Data Preprocessing

To ensure the quality of the input data for deep learning models, a comprehensive preprocessing pipeline was applied to the raw Twitter dataset. As the original comments contained significant noise, we implemented a multi-step cleaning process:

Noise Removal: Using regular expressions, we filtered out non-textual elements, including website links (URLs), user mentions (e.g., “@username”), automatic reply tags (#), and emojis, which do not contribute to semantic personality analysis.

Normalization: All textual data were converted to lowercase to maintain consistency. Non-alphabetic characters and excessive whitespace were removed.

Filtering: Samples with missing text or labels, as well as duplicate entries, were eliminated. Furthermore, comments with excessively short lengths after cleaning were removed to ensure sufficient semantic content.



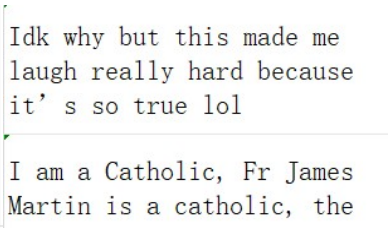


Figure 1: Text data before cleaning

Figure 2: Text data after cleaning

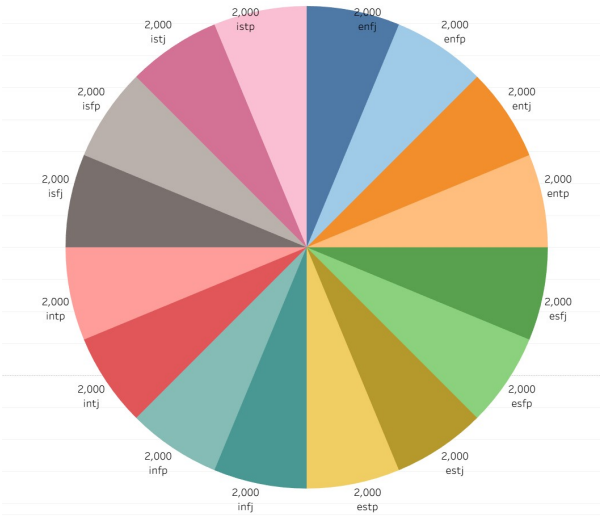


Figure 3: Data distribution after balancing (Proportion)

Following the initial cleaning, a statistical analysis revealed an imbalance in the

dataset distribution across MBTI types. To prevent the model from biasing towards majority classes, we adopted a random sampling strategy to balance the training set. Specifically, 2000 comments were selected from each of the 16 personality categories to ensure sampling equilibrium.

3.2 Feature Extraction Techniques

Before feeding data into the classification model, we conducted lexical feature extraction to understand word usage patterns.

In addition to visual inspection, we employed Term Frequency-Inverse Document Frequency (TF-IDF) to quantify term importance. This method allows us to identify keywords that are statistically significant for specific personality types by suppressing common words and highlighting rare but discriminative terms.

While TF-IDF provides a baseline for word importance, it revealed limited distinctiveness for frequent terms. To capture more subtle linguistic patterns, we employed Keyness Analysis using the quanteda package. We utilized the likelihood ratio test (G^2 statistic) to identify keywords that are statistically overrepresented in specific personality dimensions.

3.3 Model Architecture: TextCNN

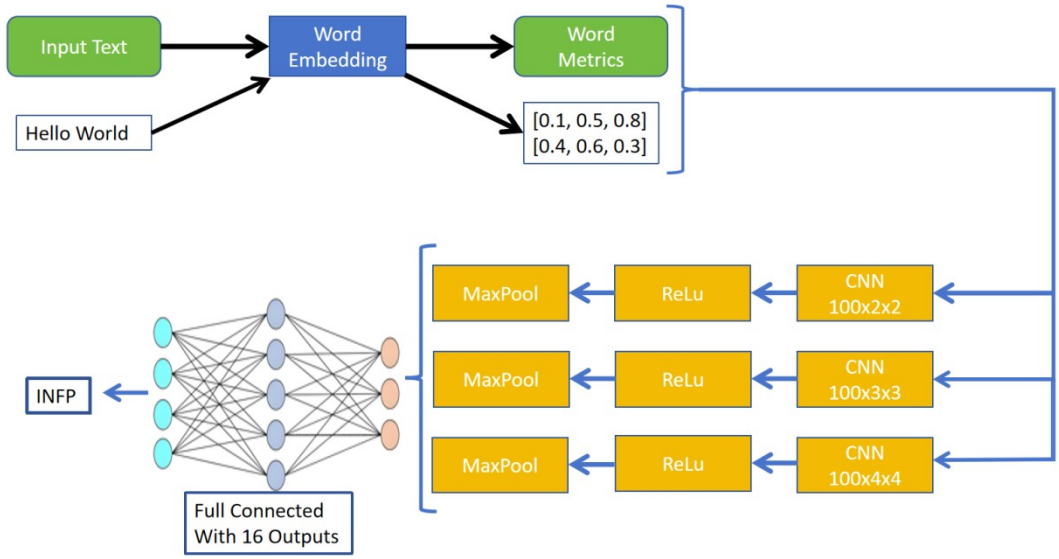


Figure 4:Architecture Diagram of the TextCNN Model

For the classification task, we implemented the TextCNN model, a specialized Convolutional Neural Network for text processing. The architecture consists of four primary components: word embeddings, convolutional layers, pooling layers, and fully connected layers.

Input Layer (Word Embedding): The model first converts the preprocessed text format into a word embedding matrix format, representing the text length and embedding dimension.

Convolutional Layer: The matrix is processed in parallel through multiple CNN layers with varying kernel sizes (e.g., 2x2, 3x3). This design allows the model to capture local semantic features and n-gram patterns effectively.

Pooling and Output: A ReLU activation function follows the convolution operation, and max-pooling is applied to extract the most significant features before passing them to the fully connected layers for final classification.

4 Experiments and Results

4.1 Exploratory Data Analysis

Prior to model training, an Exploratory Data Analysis (EDA) was conducted to assess the representativeness of our dataset. We compared the frequency of each MBTI type in our dataset against established global and Chinese population norms, as reported in the 16Personalities World Personality Map (Fourth Edition).

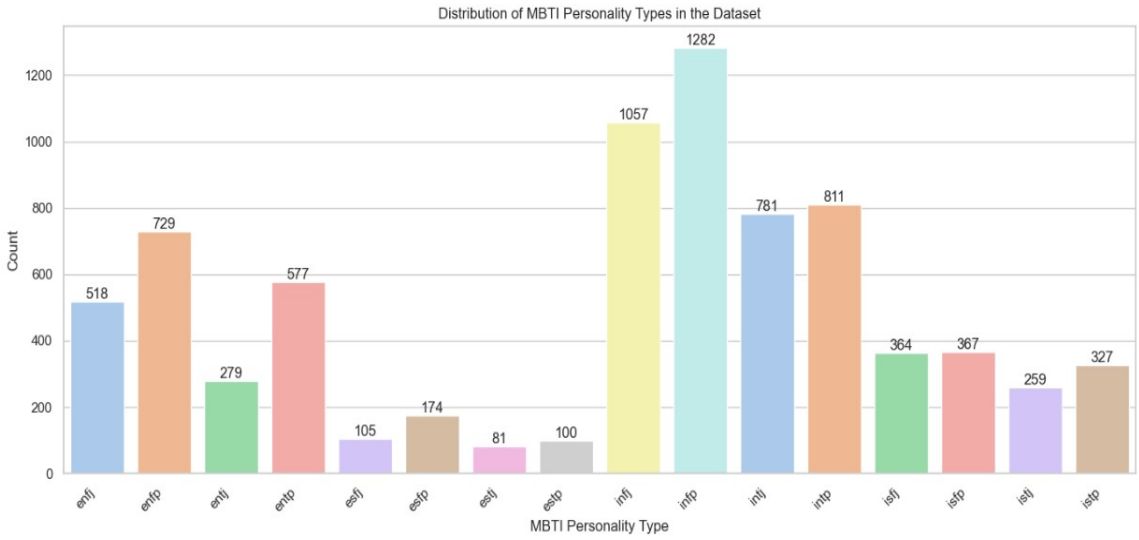


Figure 5: Comparison of MBTI type distribution

As shown in Figure 5, while ISTJ is typically the most common type globally, our social media dataset reveals a distinct divergence. Specifically, personality types in the "ES" cluster (including ESTJ, ESTP, ESFP, and ESFJ) are significantly underrepresented. This discrepancy likely reflects behavioral differences in social media usage, suggesting that ES-type individuals may be less active on Twitter or less inclined to publicly disclose their personality type.

Additionally, to ensure data quality, we analyzed the user activity levels. As illustrated in the box plot below, the median post length across all MBTI types falls consistently within the range of approximately 1,000 to 1,500 words, confirming the data's suitability for textual analysis

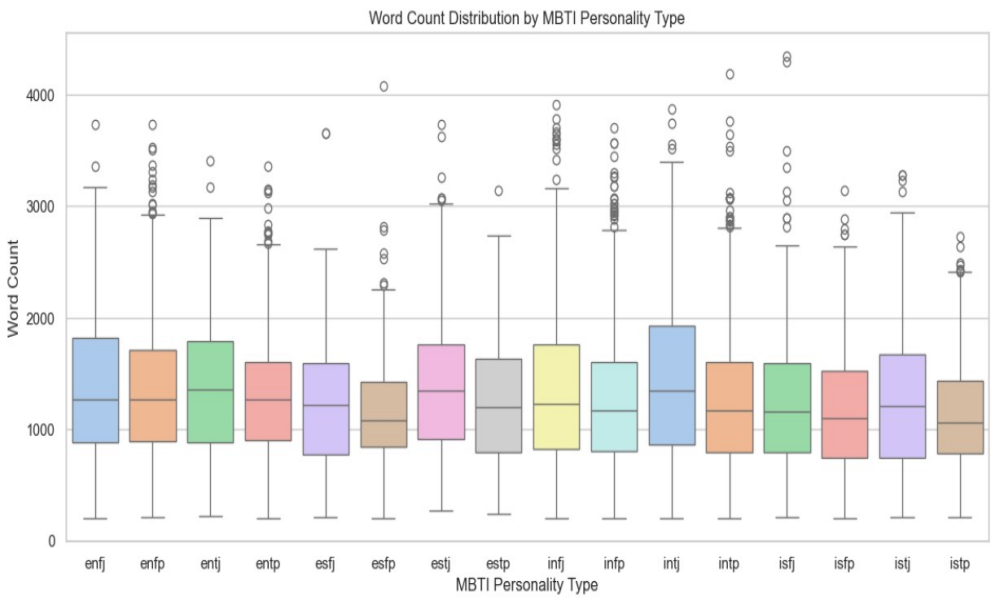


Figure 6: Box plot of average posts per user

4.2 Linguistic Analysis Results

4.2.1 Word Cloud Analysis

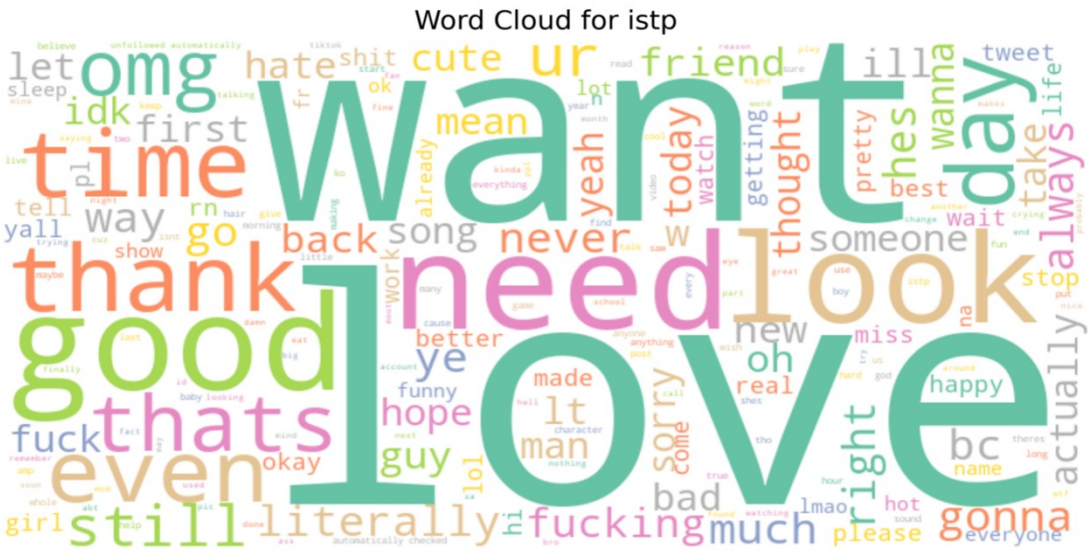


Figure 7: Word Cloud comparison (e.g., ISTP vs. ENFJ)

To visualize the linguistic preferences of different personalities, we generated word clouds. A deep cleaning process was performed to remove high-frequency but low-semantic-value words. This included standard NLTK stopwords and platform-specific terms such as "get", "thing", "twitter", and "mbti", which effectively prevented distortion in the visualization.

4.2.2 Keyness Analysis

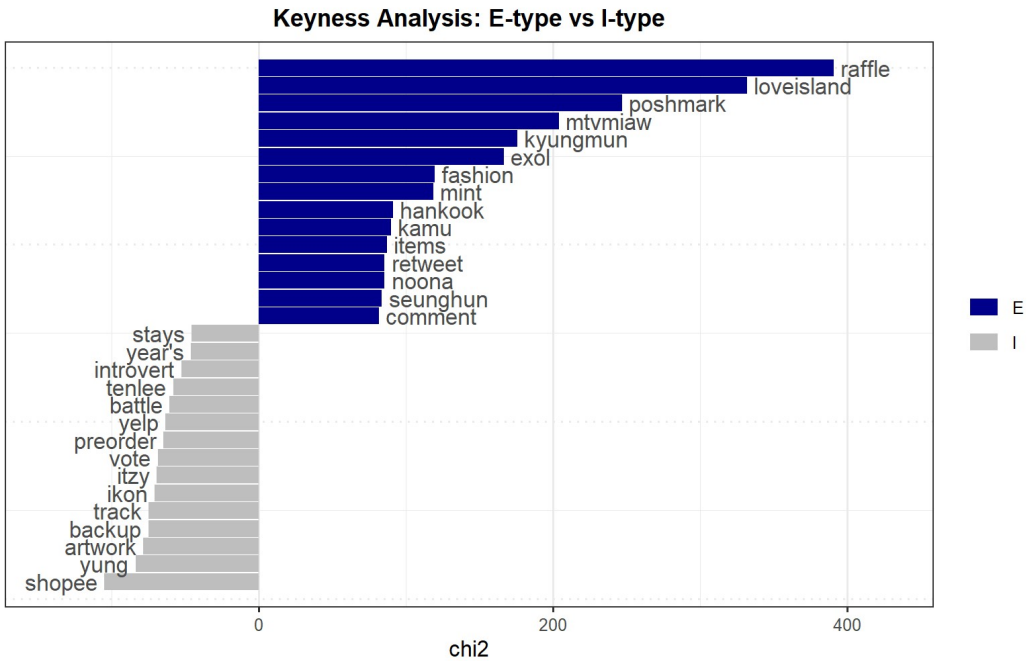


Figure 8: Top 15 keyness features for Extraversion vs. Introversion

The analysis highlights a clear dichotomy in focus. Extraverts (E) are characterized by terms related to social engagement and external events (e.g., "raffle", "loveisland"), whereas Introverts (I) exhibit a lexicon focused on internal states and specific fandom

interests (e.g., "genshin", "commission").

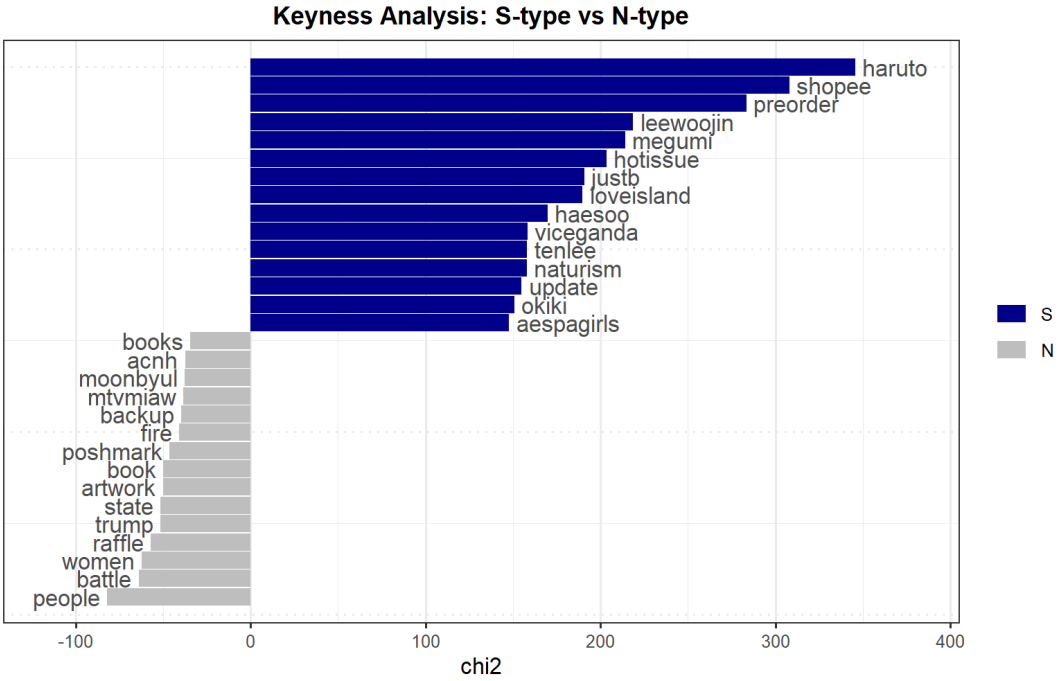


Figure 9: Top 15 keyness features for Sensing vs. Intuition

Sensing (S) types demonstrate a preference for concrete entities, often referencing specific idols or celebrities. In contrast, Intuitive (N) types show a higher usage of abstract concepts and terms related to broader societal issues (e.g., "government", "society", "women"), reflecting their theoretical orientation.

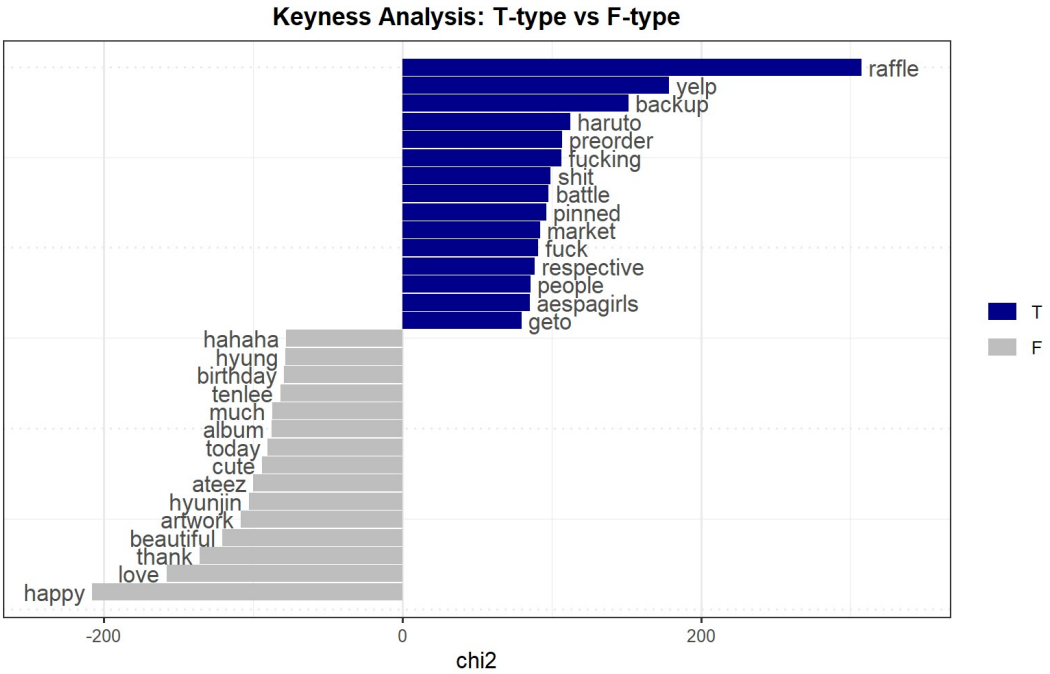


Figure 10: Top 15 keyness features for Thinking vs. Feeling

The T-F dimension reveals a contrast between transactional and aesthetic language. Thinking (T) types frequently use terms associated with efficiency and cryptocurrency (e.g., "crypto", "nft", "project"). Conversely, Feeling (F) types prioritize aesthetic appreciation and emotional expression, with keywords such as "artwork" and "beautiful" being highly prominent.

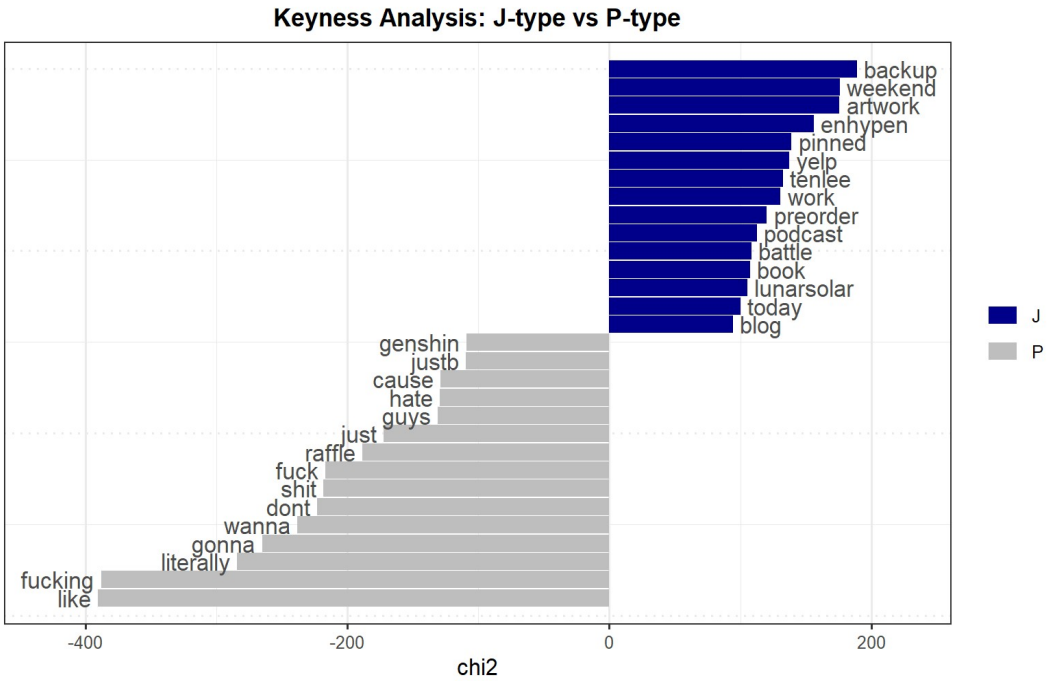


Figure 11: Top 15 keyness features for Judging vs. Perceiving

This dimension reflects a conflict between structure and spontaneity. Judging (J) types use vocabulary linked to planning and evaluation, while Perceiving (P) types favor terms related to spontaneous rewards and gaming (e.g., "raffle", "game"), indicating a more flexible lifestyle.

4.2.3 Emotional Expression Analysis

Before evaluating the predictive models, we analyzed the linguistic features associated with different personality dimensions.

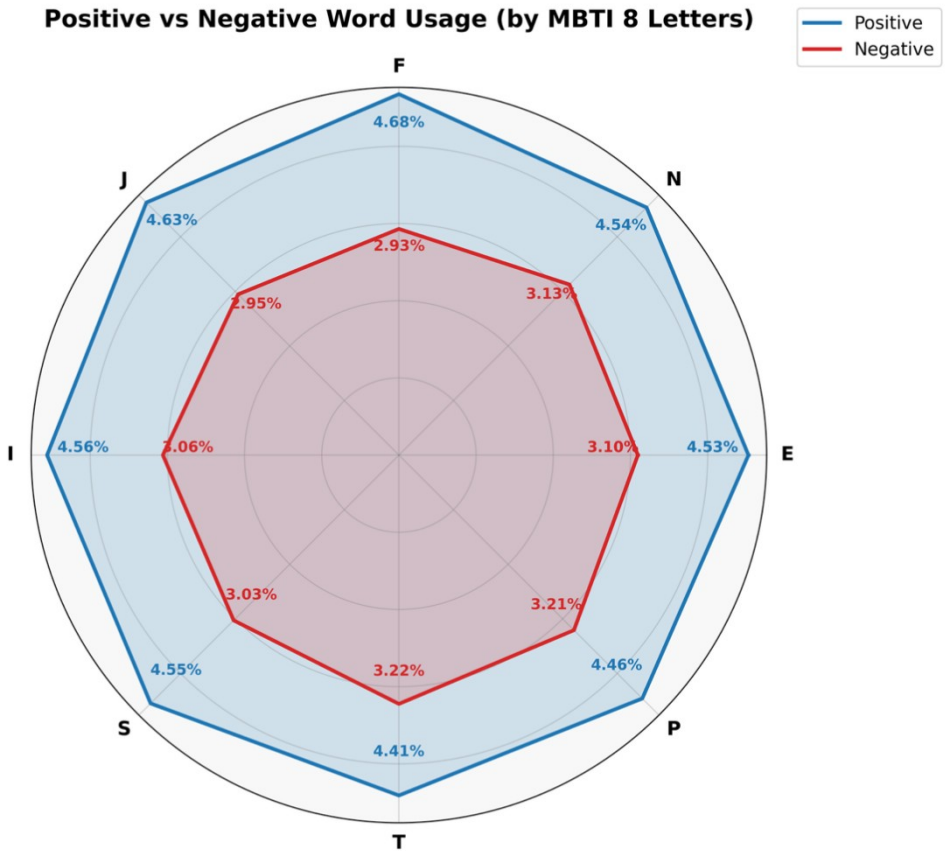


Figure 12: Emotional word usage across MBTI dimensions.

Using lexicon-based matching, we evaluated the emotional polarity of different MBTI types. The results, visualized via radar charts, indicate a general positivity bias across all personality types, with positive word usage (approx. 4.4% – 4.7%) consistently exceeding negative usage (approx. 2.9% – 3.2%). Specifically, "Feeling" (F) and "Extraverted" (E) types demonstrated the highest frequency of positive vocabulary, aligning with their psychological tendency towards social harmony and expressiveness. In contrast, "Thinking" (T) and "Intuitive" (N) types showed a slightly higher usage of negative emotion words compared to their counterparts, potentially reflecting a more critical or analytical cognitive style.

4.2.4 Cross-Dimensional Keyness Discussion

Beyond single dimensions, our TF-IDF analysis revealed terms with significance across multiple dimensions. For instance, the term "raffle" showed high keyness in both

Extraversion and Perceiving dimensions, suggesting a link to spontaneous, experience-seeking behavior. Similarly, "artwork" was characteristic of both Intuition and Feeling types, consistent with theoretical alignments between intuitive meaning-making and aesthetic appreciation.

4.3 Predictive Model Performance

4.3.1 TextCNN Evaluation

We evaluated the performance of the TextCNN model on the processed dataset. The model demonstrated robust capabilities in extracting semantic features from short texts. On the filtered test set, which contains rich MBTI-related information, the TextCNN model achieved an accuracy rate of approximately 80%.

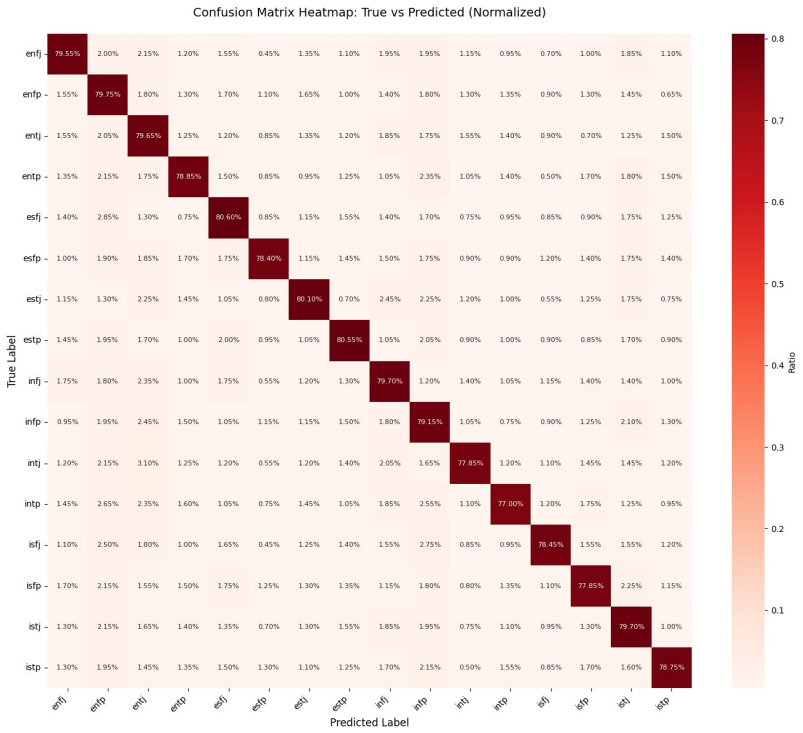


Figure 13: Classification accuracy of the TextCNN model across different personality categories

4.3.2 Robustness and Limitations

Despite the high peak accuracy, our experiments revealed significant sensitivity to data quality. When tested on noisy data containing irrelevant names or groups, the model's prediction accuracy dropped significantly to approximately 10%. Furthermore,

the model exhibited signs of overfitting during training; while training loss decreased steadily, validation accuracy occasionally fluctuated, dropping below 40% in severe overfitting scenarios. This highlights the critical importance of the rigorous data cleaning and balancing procedures described in Section 3.1.

4.3.3 Confusion Matrix Analysis

To further diagnose classification errors, we analyzed the confusion matrix. The confusion matrix reveals that while the model distinguishes well between distinct types like INTJ and ESFP, it faces challenges with types sharing dominant cognitive functions (e.g., INTP vs. INTJ). This suggests that while syntactic patterns are captured effectively, subtle semantic distinctions between similar personalities remain a challenge for the current architecture.

5 Conclusion and Future Work

5.1 Conclusion

This study establishes a systematic framework for predicting MBTI personality types from Twitter text, integrating both interpretative linguistic analysis and deep learning classification. Our findings validate that digital footprints on social media serve as reliable markers for psychological traits.

From a linguistic perspective, we identified distinctive vocabulary patterns associated with each MBTI dimension. The keyness analysis revealed that Extraversion (E) and Perceiving (P) types are linked to spontaneous, experience-seeking terms (e.g., "raffle"), while Intuition (N) and Feeling (F) types align with aesthetic and interpretative language (e.g., "artwork"). Furthermore, our sentiment analysis uncovered a general positivity bias across all types, with "Feeling" types exhibiting the highest frequency of positive emotional expression, reflecting their tendency towards social harmony.

In terms of predictive modeling, the TextCNN architecture proved effective in extracting semantic features from user comments. Under specific filtered conditions rich in personality information, the model achieved an accuracy rate of approximately 80%. This demonstrates that Convolutional Neural Networks can successfully capture the local syntactic patterns indicative of personality types.

5.2 Limitations and Future Work

Despite these promising results, several limitations remain. First, the dataset exhibits a significant demographic skew, with an underrepresentation of "ES" cluster types (ESTJ, ESTP, ESFP, ESFJ) compared to global population norms, which may introduce sampling bias. Second, the TextCNN model showed limited generalization capabilities and high sensitivity to noise. The accuracy dropped to approximately 10% when processing data containing irrelevant names or groups, and the model was prone to overfitting during training.

Future research will focus on optimizing model robustness to noise and improving generalization performance on small-sample categories. Additionally, incorporating

more diverse datasets could help mitigate the distribution imbalance, further enhancing the practical applicability of personality computing in computational social science.